Problem 4 Handout II
CNLF Finite-Difference Scheme for Viscous Burgers' Equation

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1 Conservative Form

The viscous Burgers' equation is

$$u_t + uu_x = \varepsilon u_{xx}. \tag{1}$$

Since

$$uu_x = \frac{1}{2}(u^2)_x, \tag{2}$$

we use the conservative form

$$u_t + \frac{1}{2}(u^2)_x = \varepsilon u_{xx}. \tag{3}$$

This form is important because the nonlinear term is treated as the derivative of a flux:

$$f(u) = \frac{u^2}{2}. \tag{4}$$

For nonlinear conservation laws, discretising the flux difference is usually more robust than discretising uu_x directly.

2 Grid and Notation

Let

$$x_m = x_L + mh, \qquad t_n = nk, \qquad V_m^n \approx u(x_m, t_n). \tag{5}$$

Define the second-difference operator

$$\delta_x^2 V_m^n = V_{m+1}^n - 2V_m^n + V_{m-1}^n. \tag{6}$$

The CNLF method combines three choices:

- leap-frog in time for u_t;
- an explicit centred conservative difference for $(u^2)_x/2$;
- Crank–Nicolson averaging for the diffusion term εu_{xx}.

3 Derivation of the CNLF Scheme

Approximate the time derivative at (x_m, t_n) by

$$u_t(x_m, t_n) \approx \frac{V_m^{n+1} - V_m^{n-1}}{2k}. \tag{7}$$

Approximate the conservative nonlinear flux by

$$\frac{1}{2}(u^2)_x(x_m, t_n) \approx \frac{1}{2}\frac{(V_{m+1}^n)^2 - (V_{m-1}^n)^2}{2h} = \frac{(V_{m+1}^n)^2 - (V_{m-1}^n)^2}{4h}. \tag{8}$$

Finally, approximate diffusion by a Crank–Nicolson average over the two leap-frog levels:

$$u_{xx}(x_m, t_n) \approx \frac{1}{2}\left(\frac{\delta_x^2 V_m^{n+1}}{h^2} + \frac{\delta_x^2 V_m^{n-1}}{h^2}\right). \tag{9}$$

Substituting these into (3) gives

$$\frac{V_m^{n+1} - V_m^{n-1}}{2k} + \frac{(V_{m+1}^n)^2 - (V_{m-1}^n)^2}{4h} = \frac{\varepsilon}{2h^2}\left(\delta_x^2 V_m^{n+1} + \delta_x^2 V_m^{n-1}\right). \tag{10}$$

Multiplying by $2k$ and setting

$$\mu = \frac{\varepsilon k}{h^2}, \tag{11}$$

we obtain the practical update

$$V_m^{n+1} - \mu\delta_x^2 V_m^{n+1} = V_m^{n-1} + \mu\delta_x^2 V_m^{n-1} - \frac{k}{2h}\left((V_{m+1}^n)^2 - (V_{m-1}^n)^2\right). \tag{12}$$

4 Linear System at Each Step

Although Burgers' equation is nonlinear, the CNLF update is linear in the new unknowns V^{n+1}. The nonlinear flux is evaluated at time level n, so it is already known when solving for V^{n+1}.

Expanding the left-hand side of (12),

$$V_m^{n+1} - \mu\left(V_{m+1}^{n+1} - 2V_m^{n+1} + V_{m-1}^{n+1}\right) = -\mu V_{m-1}^{n+1} + (1+2\mu)V_m^{n+1} - \mu V_{m+1}^{n+1}. \tag{13}$$

Thus, for interior nodes, the new solution is obtained from a tridiagonal linear system with coefficients

$$\text{subdiagonal} = -\mu, \qquad \text{diagonal} = 1 + 2\mu, \qquad \text{superdiagonal} = -\mu. \tag{14}$$

Known boundary values at the new time are moved to the right-hand side. This is why the travelling-wave test can impose exact boundary values from the exact solution.

5 Two-Step Startup

CNLF is a two-step method. To compute V^{n+1}, it needs both V^n and V^{n-1}. Therefore a computation must provide

$$V^0 \approx u(\cdot, 0), \qquad V^1 \approx u(\cdot, k). \tag{15}$$

For the travelling-wave convergence tests, the notebook uses the exact travelling wave to set V^1. This isolates the accuracy of the CNLF update itself. For a general problem, one may instead compute V^1 with a one-step Crank–Nicolson startup.

6 Error Norms and Observed Rates

For a final time $T = t_N$, define the grid error

$$e_m^N = V_m^N - u(x_m, T). \tag{16}$$

The maximum norm and discrete L^2 norm used in the tables are

$$\left\|e^N\right\|_\infty = \max_m |e_m^N|, \qquad \left\|e^N\right\|_2 = \left(h \sum_m |e_m^N|^2\right)^{1/2}. \tag{17}$$

For two successive errors E_h and $E_{h/2}$, the observed rate is

$$\mathrm{EOC} = \frac{\log(E_h/E_{h/2})}{\log 2}. \tag{18}$$

7 Local Truncation Error

Let $u_m^n = u(x_m, t_n)$ be the exact smooth solution sampled on the grid, and let

$$q(x,t) = u(x,t)^2. \tag{19}$$

The leap-frog time difference satisfies

$$\frac{u_m^{n+1} - u_m^{n-1}}{2k} = u_t + \frac{k^2}{6} u_{ttt} + \mathcal{O}(k^4). \tag{20}$$

The conservative nonlinear flux satisfies

$$\frac{(u_{m+1}^n)^2 - (u_{m-1}^n)^2}{4h} = \frac{1}{2}\frac{q(x_m + h, t_n) - q(x_m - h, t_n)}{2h} \tag{21}$$

$$= \frac{1}{2} q_x + \frac{h^2}{12} q_{xxx} + \mathcal{O}(h^4) \tag{22}$$

$$= \frac{1}{2}(u^2)_x + \frac{h^2}{12}(u^2)_{xxx} + \mathcal{O}(h^4). \tag{23}$$

For the diffusion term,

$$\frac{\delta_x^2 u_m^{n\pm1}}{h^2} = u_{xx}(x_m, t_{n\pm1}) + \frac{h^2}{12} u_{xxxx}(x_m, t_{n\pm1}) + \mathcal{O}(h^4). \tag{24}$$

Averaging the two time levels gives

$$\frac{\delta_x^2 u_m^{n+1} + \delta_x^2 u_m^{n-1}}{2h^2} = u_{xx} + \frac{k^2}{2} u_{xxtt} + \frac{h^2}{12} u_{xxxx} + \mathcal{O}(k^4 + k^2h^2 + h^4). \tag{25}$$

Substituting the exact solution into the discrete residual, and using (3), gives

$$\begin{aligned} \tau_m^n = \frac{k^2}{6} u_{ttt} + \frac{h^2}{12}(u^2)_{xxx} - \varepsilon\left(\frac{k^2}{2} u_{xxtt} + \frac{h^2}{12} u_{xxxx}\right) \\ + \mathcal{O}(k^4 + k^2h^2 + h^4). \end{aligned} \tag{26}$$

Therefore

$$\boxed{\tau_m^n = \mathcal{O}(k^2 + h^2).} \tag{27}$$

This is a smooth-solution consistency statement. With a second-order startup and a stable computation, one expects second-order global convergence.

8 Travelling-Wave Convergence Data

The manufactured exact solution is the travelling wave with

$$\omega = 0.5, \qquad \varepsilon = 0.1, \qquad x_0 = 3, \qquad -5 \le x \le 5, \qquad T = 2. \tag{28}$$

For the high-resolution run with $h = k = 10^{-3}$, the notebook reports

$$\|e(T)\|_\infty = 4.548 \times 10^{-7}. \tag{29}$$

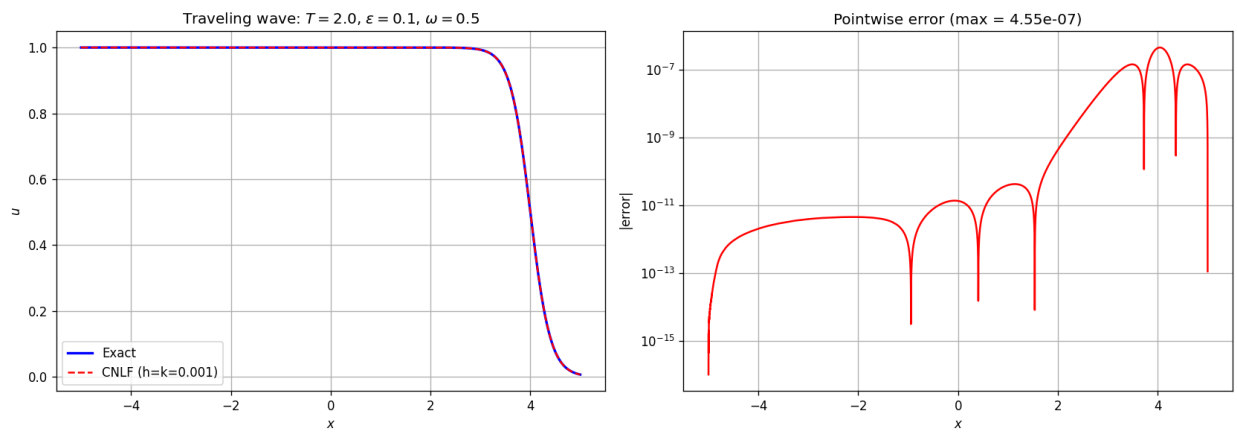

Figure 1: CNLF travelling-wave computation and pointwise absolute error.

Table 1: Coupled convergence with $h = k$.

$h = k$	$\|e(T)\|_\infty$	rate
0.0400	6.056×10^{-4}	–
0.0200	1.513×10^{-4}	2.001
0.0100	3.790×10^{-5}	1.997
0.0050	9.485×10^{-6}	1.998

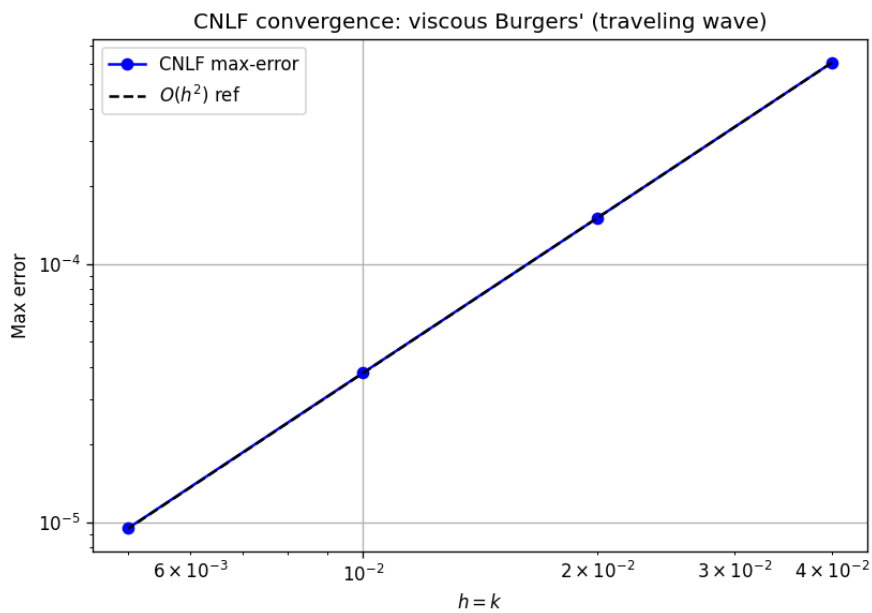

Figure 2: Coupled second-order convergence for the CNLF finite-difference method.

9 Separate Time and Space Convergence

To isolate time accuracy, the spatial mesh is made very fine:

$$h = 2.5 \times 10^{-4}. \tag{30}$$

Then the measured L^2 error behaves as $O(k^2)$.

Table 2: Temporal convergence with $h = 2.5 \times 10^{-4}$.

k	$\|e\|_2$	rate
8.0×10^{-3}	3.663×10^{-6}	–
4.0×10^{-3}	9.125×10^{-7}	2.005
2.0×10^{-3}	2.250×10^{-7}	2.020
1.0×10^{-3}	5.316×10^{-8}	2.081

To isolate space accuracy, the time step is made very small:

$$k = 10^{-4}. \tag{31}$$

Then the measured L^2 error behaves as $O(h^2)$.

Table 3: Spatial convergence with $k = 10^{-4}$.

h	$\|e\|_2$	rate
4.0×10^{-2}	1.584×10^{-4}	–
2.0×10^{-2}	3.963×10^{-5}	1.999
1.0×10^{-2}	9.911×10^{-6}	1.999
5.0×10^{-3}	2.478×10^{-6}	2.000

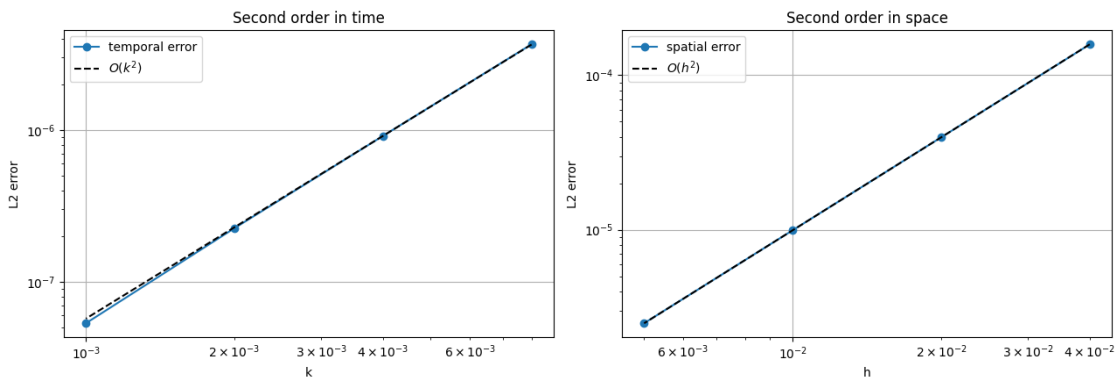

Figure 3: Separate temporal and spatial convergence tests.

10 Conclusion

The finite-difference method is second order because each of its three components is second order in the appropriate sense: leap-frog is second order in time, the centred flux difference is second order in space, and the diffusion term uses a Crank–Nicolson average with a centred second difference. The convergence tables confirm the expected $\mathcal{O}(k^2 + h^2)$ behaviour.